

21 L reacts with excess hot concentrated acidified KMnO_4 to produce three different organic molecules, $\text{CH}_3\text{CO}_2\text{H}$, $(\text{CH}_3)_2\text{CO}$ and $\text{HO}_2\text{CCOCH}_3$. Which statements are correct?

- 1 L can be 2,4-dimethylhexa-2,4-diene or 2,3-dimethylhexa-2,4-diene.
- 2 The three products formed can be distinguished from one another by using $\text{Na}_2\text{CO}_3(\text{s})$ followed by 2,4-DNPH.
- 3 L can also react with excess hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to produce the same three organic molecules.

- A** 1, 2 and 3
- B** 1 and 2 only
- C** 2 and 3 only
- D** 1 only